



The Compatibility of a Boron-Based Anion Receptor with the Carbon Anode in Lithium-Ion Batteries

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The electrochemical stability of the boron anion receptor, tris(pentafluorophenyl) borane (TPFPB), on a carbonaceous mesocarbon microbeads (MCMB) electrode and its compatibility with the passivated solid electrolyte interphase (SEI) layer on the carbon anode were investigated. Comparison of the irreversible capacity loss of the MCMB electrode during initial galvanostatic cycling using electrolyte with and without the TPFPB additive indicates excellent electrochemical stability of TPFPB on the carbonaceous electrode. Cyclic voltammetry studies show that the SEI layer on the surface of the carbon electrode can be formed through the decomposition of ethylene carbonate in the presence of TPFPB. Prolonged cycling test verifies the long-term stability of the SEI layer on carbon in the presence of the TPFPB additive. The SEI layer is not dissolved by TPFPB even after heat-treatment under conditions which dissolve LiF salt. This suggested a cross-linked structure for the SEI layer on carbon electrode. A Li-ion cell using an electrolyte containing TPFPB displays better cycling performance than a cell without TPFPB under the same conditions. © 2003 The Electrochemical Society. [DOI: 10.1149/1.1536475] All rights reserved.

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The low conductivity of nonaqueous electrolyte for lithium batteries is mainly due to ion pairing or low solubility of the lithium salts in organic solvents. Development of neutral ligands as additives is a very effective way to break down ion pairing and enhance the ionic conductivity.¹⁻⁵ Anion coordination ligands are much more favorable than cation coordination ligands for a lithium battery electrolyte, because they increase both ionic conductivity and lithium transference number which are necessary to achieve high power density and good rechargeability for lithium batteries.⁶ Anion receptors are a very active field of research with most of the work aimed at molecular recognition to mimic how ion-binding protons control ion transport in biological membranes. Most anion receptors are based either on positively charged sites, hydrogen bonding, or Lewis acid metal centers. None of these are suitable for use in nonaqueous electrolytes. A new family of boron-based anion complexing agents has been synthesized and reported by our group recently. These compounds are based on electron deficient borate or borane compounds with various fluorinated aryl or alkyl groups.⁷⁻¹⁰ Some of these boron anion receptors can dramatically promote the dissolution and conductivity of various lithium salts in organic solvents. Even LiF, which is normally insoluble in organic solvents, can be dissolved in several organic solvents with the assistance of these anion receptors. The additives increase the solubility of LiF from $<10^{-5}$ to >1.0 M and conductivities of $\sim 10^{-3}$ S/cm can be achieved in carbonate solvents. Most of these boron-based anion acceptors have a high solubility in conventional nonaqueous solvents. Several of these compounds have exceptional chemical, electrochemical, and thermal stability. Composite electrolytes made with boron anion receptors and LiF salt or $\text{CF}_3\text{CO}_2\text{Li}$ salts have been verified as having high stability and compatibility with LiMn_2O_4 , $\text{LiNi}_x\text{Co}_{1-x}\text{O}_2$ cathodes. LiMn_2O_4 spinel also displays much better cycling performance at elevated temperatures when using these nonacidic composite electrolytes instead of LiPF_6 -based electrolytes.¹¹ More recently, one of the boron anion receptors, tris(pentafluorophenyl) borane (TPFPB), was found to significantly improve the thermal stability of LiPF_6 -based electrolytes.¹² Therefore, boron-based anion receptors have displayed promising application in lithium battery technologies.

However, the stability of boron anion receptors with the carbonaceous anode has not been investigated. It is commonly agreed that the passivated solid electrolyte interphase (SEI) layer on the surface of a carbon electrode is formed via the decomposition of electrolyte components. The resulting film blocks further electrolyte reduction

while still allowing lithium-ion transport.^{13,14} The formation of a stable passivated SEI film is crucial for the use of carbon electrodes in lithium batteries. The major components of SEI layer are believed to be various lithium salts, such as lithium carbonate, lithium alkyl carbonate, and lithium alkoxide.¹⁵ Due to the strong anion coordination effect of the boron compounds, there is concern that boron anion receptors may degrade the SEI layer on the carbon electrode by dissolving these lithium salts and exposing the carbon surface to the electrolyte. This paper summarizes our preliminary results about the compatibility of the boron compound TPFPB with the carbon anode and explores the application of boron anion receptors in Li-ion batteries.

Experimental

Materials.—TPFPB compound was synthesized as described in our previous work.⁷ The commercial electrolyte, 1 M LiPF_6 in ethylene carbonate/dimethyl carbonate (EC/DMC, 1:1 volume ratio), was purchased from EM Industries Inc. High rate carbonaceous mesocarbon microbeads (MCMB) anodes and $\text{LiNi}_{0.80}\text{Co}_{0.15}\text{Al}_{0.05}\text{O}_2$ cathode were provided by Quallion LLC.

Electrochemical characterization.—Electrochemical characterization was performed in a standard coin cell configuration. Galvanostatic cycling was carried out on an Arbin battery test system and the cells were cycled under a constant current density of 0.12 mA/cm^2 at room temperature. A Solatron SI 1287 electrochemical interface system was used to perform cyclic voltammetry (CV) experiments under software control. The scan rate was 0.1 mV/s .

Results and Discussion

The compatibility of the boron anion receptor with carbon anode was studied by adding 0.1 M TPFPB to a commercial electrolyte of 1 M LiPF_6 /EC:DMC (1:1) (EM Industry Inc.). Cyclic voltammogram studies were performed at the MCMB electrode in 1 M LiPF_6 /EC:DMC (1:1) electrolyte with and without 0.1 M TPFPB. The scan rate is 0.1 mV/s and voltage range was 3 to 0 V (vs. Li/Li^+). Figure 1 compares the cyclic voltammogram from the first scan. The cell with the TPFPB additive exhibits an almost identical CV profile as the one without the TPFPB additive during the first cycle. Preliminary observation identified only one reduction peak at 0.64 V before the Li intercalation reaction in both cases. The reduction peak at 0.64 V is induced by the decomposition of the EC solvent.¹⁵ Figure 2 shows an expansion of the current scale in Fig. 1 in the voltage range of 3 to 1 V . Figure 2a shows the cathodic sweep and Fig. 2b shows the anodic sweep. The figures with an expanded current scale reveal three additional reduction peaks (1.6 , 1.9 , and 2.5 V) in the cell with the TPFPB additive. Since the current in-

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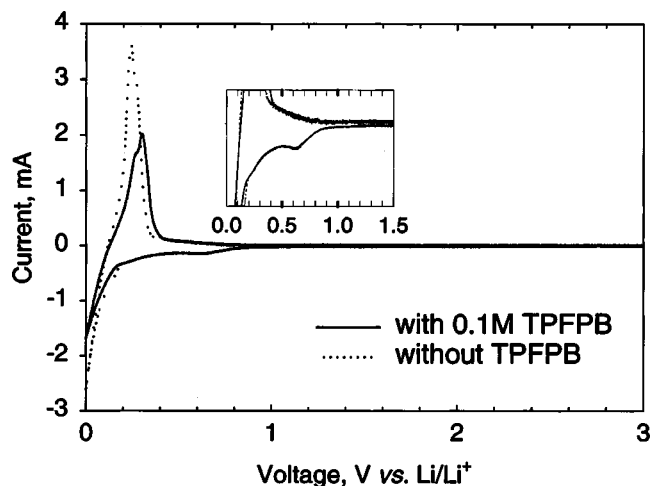


Figure 1. Cyclic voltammogram of MCMB electrode in a $\text{LiPF}_6/\text{EC}/\text{DMC}$ (1:1) electrolyte with or without 0.1 M TFPFB additive during the first scan. Inset shows detail in 0 to 1.5 V region with expanded current scale. Sweep rate, 0.1 mV/s.

duced by these reduction reactions is quite low (<7% of the current generated by EC decomposition), we postulate that they may come from an impurity in the TFPFB additive. These reduction reactions are irreversible, since no corresponding oxidation peak was observed on the anodic sweep in Fig. 2b. Figure 3 shows a comparison of CV scans on the second cycle. The peak for the EC decomposition (indicated by the reduction peak at 0.64 V) in both cases disappeared on the second cycle. Expansion of the current scale in the

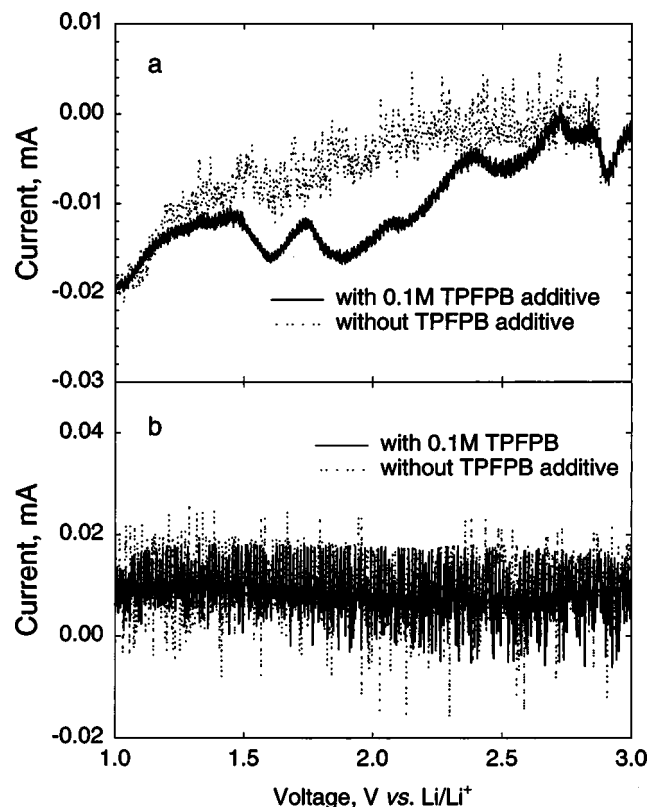


Figure 2. CV profiles with expanded current scale in the voltage range 3 to 1 V during the first scan: (a) cathodic sweep, (b) anodic sweep. Sweep rate, 0.1 mV/s.

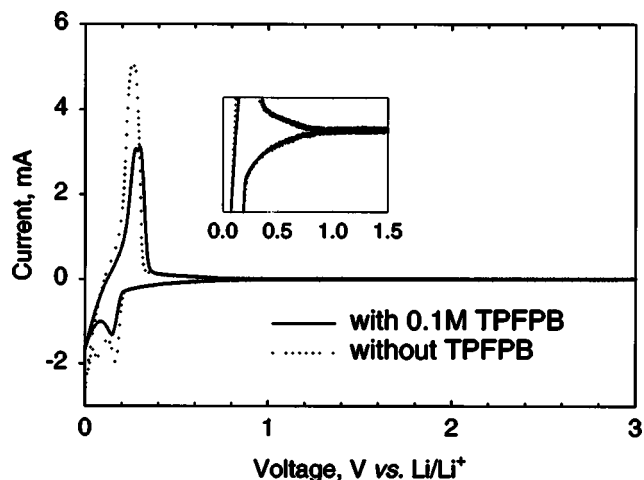


Figure 3. Cyclic voltammogram of MCMB electrode in a $\text{LiPF}_6/\text{EC}/\text{DMC}$ (1:1) electrolyte with or without a 0.1 M TFPFB additive during the second scan. Sweep rate, 0.1 mV/s.

voltage range 1 to 3 V did not reveal any additional reduction peaks besides the lithium intercalation (Fig. 4). This means that, even in the presence of TFPFB additive, a stable SEI layer has been formed without interference from the additive. Apparently, the SEI layer on the carbonaceous anode could be properly formed by decomposition of EC solvent in the presence of TFPFB. These results provide clear evidence that TFPFB is highly stable and compatible with the carbonaceous anode.

The voltage/capacity profiles of the cells with and without the TFPFB additive under constant current conditions (C/28 rate) are compared in Fig. 5. The initial irreversible capacity for the cell having the TFPFB additive is 65 mAh/g, which is almost identical to that of the cell without the TFPFB additive (63 mAh/g). Since the initial irreversible capacity is caused by decomposition of electrolyte components and irreversible intercalation of lithium, the identical initial capacity loss with and without the TFPFB implies that TFPFB does not have significant side reaction at the surface of the carbon electrode during lithium intercalation and extraction. This result is consistent with our CV observations.

Long-term stability of TFPFB with the carbon electrode was investigated by prolonged cycling testing of MCMB-Li cells. The cells using electrolyte with and without the TFPFB additive in 1 M $\text{LiPF}_6/\text{EC}/\text{DMC}$ (1:1) were cycled at room temperature at the C/12 rate. Figure 6a shows the deintercalation capacity vs. cycling num-

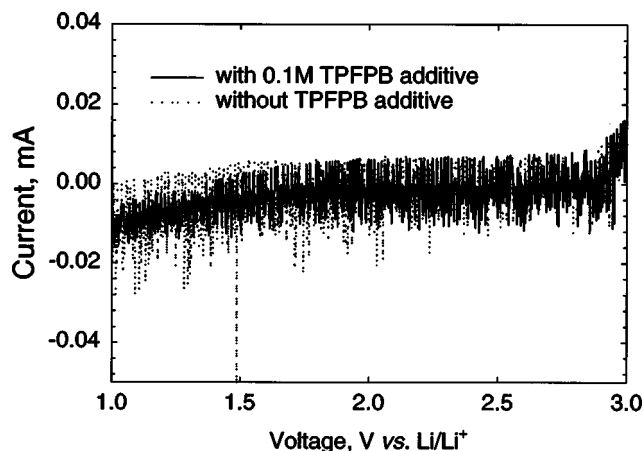


Figure 4. CV profiles with an expanded current scale in the voltage range 3 to 1 V during the second cathodic scan. Sweep rate, 0.1 mV/s.

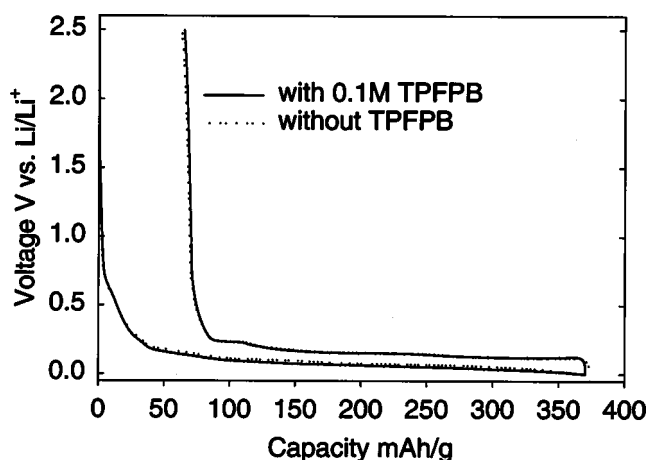


Figure 5. Voltage/capacity profiles of MCMB/Li cells with $\text{LiPF}_6/\text{EC}/\text{DMC}$ (1:1) electrolytes with or without 0.1 M TPFPB additive cycled between 0 and 2.5 V cycled under a constant current density of $0.05 \text{ mA}/\text{cm}^2$ (C/28 rate).

ber. The capacity of the cell with TPFPB additive declines during first five cycles and then starts to climb and exceed the capacity of the cell without TPFPB at about the 20th cycle. After that, the cell with the TPFPB delivered higher capacity than the cell without the TPFPB. The coulombic efficiencies of these cells during multiple cycling are plotted in Fig. 6b. Even during the first 20 cycles, when the cell with the TPFPB delivered lower capacity than the cell without the TPFPB additive, the cycling efficiencies of the cell with

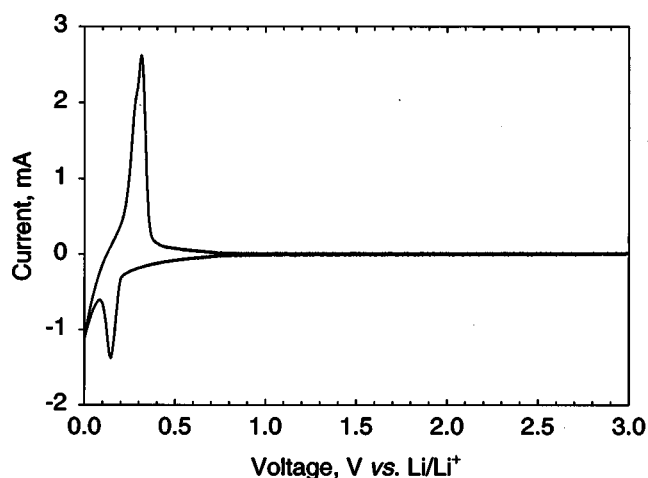


Figure 7. Cyclic voltammogram of MCMB electrode in a $\text{LiPF}_6/\text{EC}/\text{DMC}$ (1:1) electrolyte with 0.1 M TPFPB additive collected after the cell had undergone galvanostatic formation cycle at C/12 rate and then was stored at 50°C for 24 h. Sweep rate, $0.1 \text{ mV}/\text{s}$.

TPFPB are as high as the cell without TPFPB additive. These indicate excellent long-term stability of carbon electrode in the presence of TPFPB additive.

To further evaluate the stability of the SEI layer on the carbon electrode with the TPFPB additive, a MCMB/Li cell containing 0.1 M TPFPB in 1 M $\text{LiPF}_6/\text{EC}/\text{DMC}$ (1:1) electrolyte was stored in

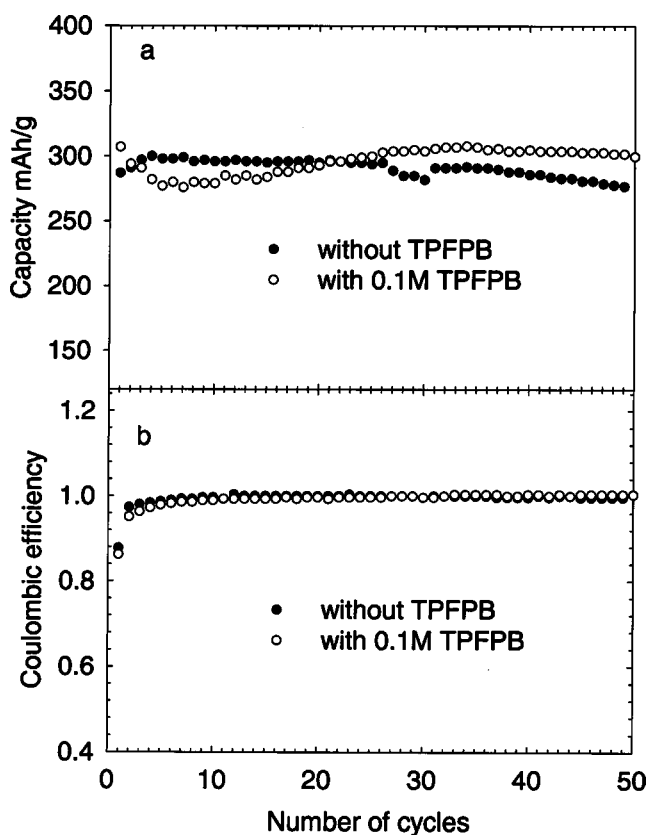


Figure 6. Cycling performance (a) and coulombic efficiency (b) of the MCMB/Li cells with a $\text{LiPF}_6/\text{EC}/\text{DMC}$ (1:1) electrolytes with or without 0.1 M TPFPB additive cycled under a constant current density of $0.12 \text{ mA}/\text{cm}^2$ (C/12 rate).

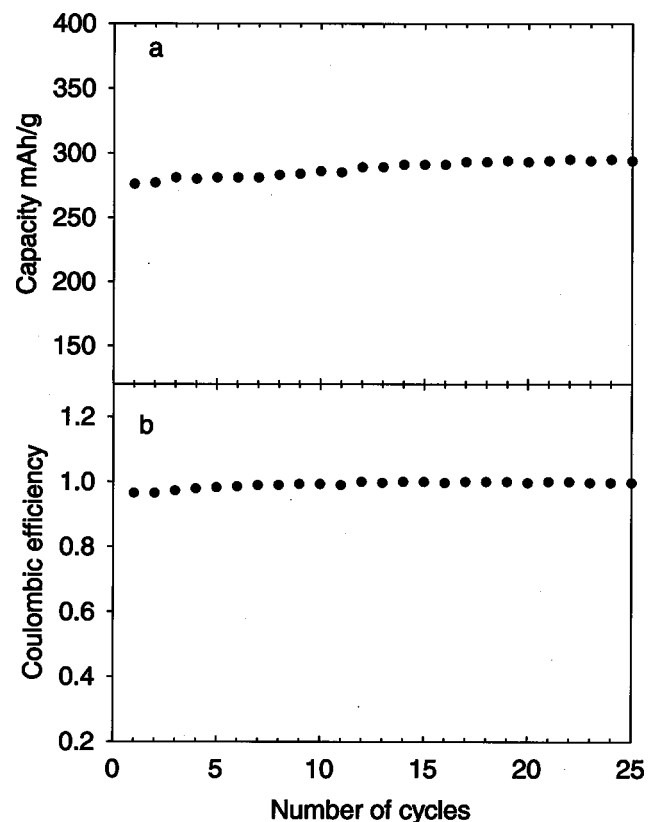


Figure 8. Cycling performance (a) and coulombic efficiency (b) of the MCMB/Li cells with a $\text{LiPF}_6/\text{EC}/\text{DMC}$ (1:1) electrolyte with 0.1 M TPFPB additive under constant current conditions ($0.12 \text{ mA}/\text{cm}^2$). Before the multiple cycling, the cell already had a galvanostatic formation cycle at the C/12 rate and was stored at 50°C for 24 h after the formation cycling.

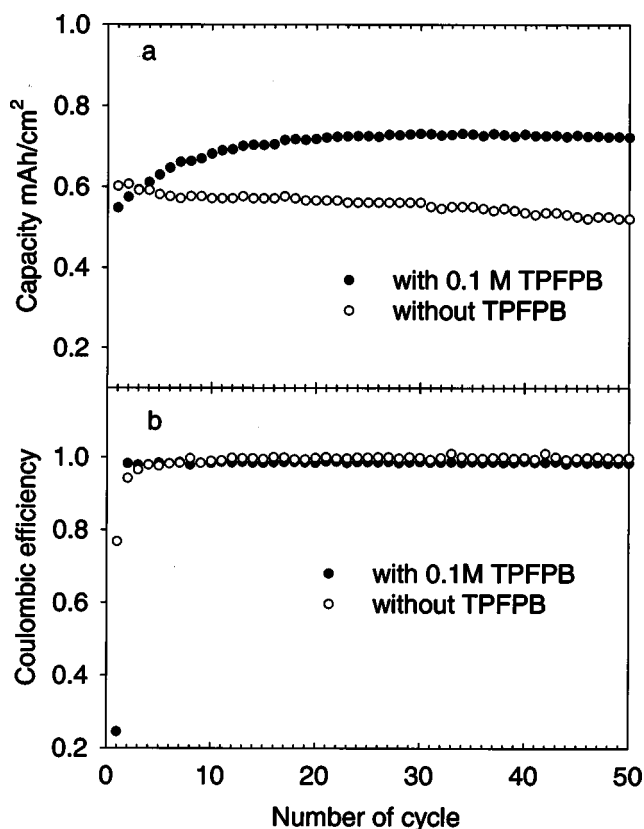


Figure 9. Cycling performance (a) and coulombic efficiency (b) of the $\text{LiNi}_{0.80}\text{Co}_{0.15}\text{Al}_{0.05}\text{O}_2/\text{MCMB}$ cells with $\text{LiPF}_6/\text{EC}/\text{DMC}$ (1:1) electrolyte with or without a 0.1 M TFPFB additive cycled under a constant current density of $0.12 \text{ mA}/\text{cm}^2$ between 2.5 and 4.2 V.

the discharge state at 50°C for 24 h after the first formation cycle at the C/12 rate. It was assumed that an SEI layer had been formed during this formation cycling. If TFPFB could dissolve the SEI layer on the carbon electrode, such heat-treatment should accelerate the dissolution process. Our previous experiments show that even LiF is dissolved in the EC-DMC solvent and reaches 1 M concentration under such conditions. After the heat-treatment, a CV of the cell was recorded to check the status of the SEI layer on the carbon electrode. As shown in Fig. 7, only the redox process corresponding to lithium intercalation and extraction was found in the scan. This indicates that the carbon anode was still well protected by the SEI layer even after such a heat-treatment in the presence of the TFPFB additive. The cell was then cycled at the C/12 rate for 20 cycles to further examine the status of the SEI layer on the carbon electrode. The deintercalation capacity and cycling efficiency are plotted in Fig. 8. After the heat-treatment, the cell delivered a slightly higher capacity and cycling efficiency than the cell without the heat-treatment. This further proves that SEI layer on carbon electrode is not attacked by the TFPFB anion receptor. Therefore, we surmise that the SEI layer may have a cross-linked structure. When it is properly formed, coordination agents will not dissolve it off from the carbon electrode.

The TFPFB additive was also tested in a Li-ion cell, $\text{LiNi}_{0.80}\text{Co}_{0.15}\text{Al}_{0.05}\text{O}_2\text{-MCMB}$. Figure 9a compares the discharge capacity of the Li-ion cells using electrolytes with and without

TPFPB additive. Except for the first few cycles, the cell with the TFPFB additive delivered higher capacity than the one without the TFPFB additive. Also, note that the discharge capacity of the cell containing TFPFB increases with cycling until the 20th cycle, and after that, the capacity remained almost constant. The cell without the TFPFB shows a decline in capacity with extended cycling. The coulombic efficiency of the cell with the TFPFB was also as high as the one without TFPFB as shown in Fig. 9b. This means that the reduced initial capacity is not due to an irreversible process. It is apparent the Li-ion cell displays better performance in the presence of the TFPFB additive. These results verified the excellent stability of TFPFB compound with both cathode and carbon anode materials and predict a promising application of TFPFB compound in lithium-ion battery. The capacity increase with cycling was observed in both the MCMB/Li half-cell and $\text{LiNi}_{0.80}\text{Co}_{0.15}\text{Al}_{0.05}\text{O}_2\text{-MCMB}$ full cell in the presence of TFPFB. The phenomenon is not understood so far and will be the subject of future study.

Conclusions

A boron anion receptor, TFPFB, displays excellent electrochemical stability with a carbonaceous MCMB electrode. CV studies show that the presence of the anion receptor TFPFB does not interfere with the formation of the passivating SEI layer on carbon electrode. CV studies show that the passivating SEI layer on the surface of carbon electrode is formed through the decomposition of EC in the presence of TFPFB. The SEI layer on the carbon electrode also remains stable in the presence of the TFPFB during prolonged cycling. The SEI layer also survived a heat-treatment in the presence of TFPFB under condition that will dissolve LiF salt. This verifies the excellent compatibility of TFPFB with the carbon electrode and suggests a cross-linked structure of the SEI layer. A Li-ion cell with TFPFB delivered higher capacity than a cell without TFPFB in prolonged cycling.

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